Bias in estimating food consumption of fish by stomach-content analysis

A. Rindorf and P. Lewy

Abstract: This study presents an analysis of the bias introduced by using simplified methods to calculate food intake of fish from stomach contents. Three sources of bias were considered: (1) the effect of estimating consumption based on a limited number of stomach samples, (2) the effect of using average contents derived from pooled stomach samples rather than individual stomachs, and (3) the effect of ignoring biological factors that affect the evacuation of prey. Estimating consumption from only two stomach samples yielded results close to the actual intake rate in a simulation study. In contrast to this, a serious positive bias was introduced by estimating food intake from the contents of pooled stomach samples. An expression is given that can be used to correct analytically for this bias. A new method, which takes into account the distribution and evacuation of individual prey types as well as the effect of other food in the stomach on evacuation, is suggested for estimating the intake of separate prey types. Simplifying the estimation by ignoring these factors biased estimates of consumption of individual prey types by up to 150% in a data example.

Résumé : Nous présentons une analyse des erreurs causées par l'utilisation de méthodes simplifiées pour calculer l'ingestion de nourriture à partir de contenus stomacaux chez les poissons. Nous avons examiné trois sources d'erreur, soit les conséquences (1) d'estimer la consommation à partir d'un petit nombre de prélèvements de contenus stomacaux, (2) d'utiliser des contenus stomacaux moyens obtenus par le regroupement de prélèvements, plutôt que des prélèvements individuels et (3) de ne pas tenir compte des facteurs biologiques qui affectent l'évacuation des proies. Dans une étude de simulation, l'estimation de la consommation à partir de seulement deux prélèvements stomacaux a donné des résultats proches de l'ingestion réelle. En revanche, l'estimation de l'ingestion de nourriture à partir de prélèvements stomacaux regroupés introduit une erreur positive importante. Nous fournissons une correction qui peut servir à rectifier cette erreur dans l'analyse. Nous suggérons une nouvelle méthode, qui tient compte de la répartition et de l'évacuation des divers types individuels de proies, ainsi que de l'effet de la présence d'autres aliments dans l'estomac sur l'évacuation, pour estimer l'ingestion particulière des différents types de proies. La simplification de l'estimation par la négligence de ces facteurs produit une erreur dans la consommation estimée de proies individuelles pouvant atteindre 150 % dans un exemple basé sur des données réelles.

[Traduit par la Rédaction]

Introduction

The interaction between predatory fish and their prey is a key subject to be addressed when attempting to understand aquatic-ecosystem dynamics. The effect of the predator on the prey depends on the link between diet composition and prey availability and on the total amount of food ingested by the predator. Both diet composition and total consumption are frequently studied by combining knowledge of evacuation rates with field studies of stomach contents. Differing methods have been suggested for this, depending on the relationship between stomach contents and evacuation rate and on the assumption made concerning the progress of feeding between consecutive sampling points (Elliott and Persson 1978; Pennington 1985; Dos Santos and Jobling 1995). The evacuation of stomach contents of a number of fish species is well described by the relationship

$$\frac{dS(t)}{dt} = -\rho S^{\alpha}(t)$$

where ρ is the evacuation rate ($g^{1-\alpha}\cdot h^{-1}$), S is stomach contents (g), and α is a unitless constant (Jones 1974; Jobling 1981; Andersen 2001). If the predator ingests food at a rate $C(t)$ at time t ($g\cdot h^{-1}$), the stomach contents vary according to the differential equation

$$\frac{dS(t)}{dt} = -\rho S^{\alpha}(t) + C(t) \quad (1)$$

This equation has been solved analytically for $\alpha = 1$ (exponential evacuation) in the case where $C(t)$ is constant over time (Eggers 1977; Elliott and Persson 1978). However, the equation does not have a general analytical solution and in many cases, food intake must be approximated numerically. In the case where α deviates from 1 and the intake rate is not necessarily constant, food intake is usually estimated by methods resembling that suggested by Pennington (1985), who estimated average hourly consumption in a sampling interval as

Received 31 July 2003. Accepted 31 August 2004. Published on the NRC Research Press Web site at http://cjfas.nrc.ca on 21 February 2005.
J17678

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 doi: 10.1139/F04-200 © 2004 NRC Canada

$$(2) \qquad C_P = \frac{\rho}{N}\sum_{i=1}^{N} E(S_i^{\alpha}) + \frac{1}{T}[E(S(T)) - E(S(0))]$$

where C_P is the average food-intake rate ($g \cdot h^{-1}$) in the sampling interval, N is the number of sampling times in the sampling interval from time 0 to T h, $E(S(0))$ and $E(S(T))$ are the average stomach contents (g) of the population sampled at 0 and T h, respectively, and $E(S_i^{\alpha})$ is the average of the stomach contents to potency α (g^{α}) in the population sampled. $E(S(0))$ and $E(S(T))$ are estimated from the average stomach contents of the fish sampled at 0 and T h, from here on denoted $\overline{S}(0)$ and $\overline{S}(T)$, whereas $E(S_i^{\alpha})$ is estimated by $\frac{1}{M_i}\sum_{h=1}^{M_i} S_{i,h}^{\alpha}$, where M_i is the number of stomachs in the ith sample and $S_{i,h}$ is the contents of the hth stomach in the ith sample, from here on denoted $\overline{S_i^{\alpha}}$ (Pennington 1985). Comparable methods have been suggested by a number of authors (Jones 1978; Armstrong 1982; Dos Santos and Jobling 1995), while others have used even simpler methods (Daan 1973; Lockwood 1980; Basimi and Grove 1985). The degree to which the estimates yielded by any of these methods are unbiased estimates of food-intake rate has hitherto only been investigated for $\alpha = 1$ (Eggers 1977; Elliott and Persson 1978; Héroux and Magnan 1996).

The calculation of consumption rate using eq. 2 in cases where $\alpha \neq 1$ requires information on the contents of single stomachs in order to estimate $E(S_i^{\alpha})$. However, the contents of all stomachs in a sample are often pooled at the time of collection to reduce the cost of examining composition. $E(S_i^{\alpha})$ is then estimated as $(\overline{S}_i)^{\alpha}$, where $\overline{S}_i$ denotes the observed average stomach contents in sample i (Jones 1978; Armstrong 1982; Patterson 1985). As can be shown using Hölder's inequality (Rao 1965), this procedure introduces a bias as $E(S_i^{\alpha}) < (E(S_i))^{\alpha}$ for $0 < \alpha < 1$. This bias has been noted by several authors (Ursin et al. 1985; Dos Santos and Jobling 1995; Andersen 2001), and is related to the proportion of stomachs that are classified as empty (Andersen 2001).

Predators with empty stomachs pose an additional problem, as these fish may be either caught in the interval between the evacuation of their last meal and the successful capture of the next prey, or not feeding for other reasons. When the average intake of the total predator population is estimated, the empty stomachs must obviously be included regardless of their origin. However, if only the intake of potentially feeding fish is desired, the individuals that are fasting for reasons other than the lack of successful capture of prey should be excluded. In the case of spawning fish, this can be done by excluding fish that are ripe or running, but generally the distinction between fasting individuals and feeding fish with empty stomachs is difficult to make, and this has led to the exclusion of empty stomachs prior to the estimation of food intake (Amundsen and Klemetsen 1988).

When the predator feeds on more than one prey species, the amount eaten of each prey type is often needed. This has been estimated by distributing the total intake among the different prey types according to the weight percentage of the prey in the total stomach contents (Armstrong 1982; Hawkins et al. 1985; Persson and Hansson 1999). The implicit assumption is that all prey types occur in the average amount in all stomachs and that the rates of evacuation of all prey types are equal. These assumptions are rarely, if ever, met (Dos Santos and Jobling 1995; Stefánsson and Pálsson 1997; Andersen 1999). Alternatively, the intake has been calculated for each prey type separately (Armstrong 1982; Patterson 1985). This allows the evacuation rate to be varied among prey types. However, the method does not take the effect of other food in the stomach into account. Calculating the intake in this way is therefore based on the assumption that only one prey type is present in each stomach or that evacuation of the prey is independent of other food in the stomach. This may hold in some cases (Magnússon and Aspelund 1997), but in many cases, each stomach contains several prey types (Rice 1988; Albert 1995) and these prey types are not evacuated independently of each other (Jones 1974; Andersen 2001). An exception to the assumption of separate evacuation of prey types is the equation given by Dos Santos and Jobling (1995). However, the parameters in their model have currently only been published for Atlantic cod (*Gadus morhua*) and furthermore, the model does not allow for changes in the average stomach contents over time.

This study presents an analysis of biases introduced by making simplifications and erroneous assumptions when estimating food intake. The investigation is divided into three parts: (1) an examination of the ability of the method suggested by Pennington (1985) to estimate food intake that occurs at a constant rate from two samples, (2) a study of the effect of using average stomach contents derived from pooled stomach samples to estimate food intake, and (3) an examination of the effect of ignoring biological factors that affect the evacuation of prey.

The difference between actual food intake and the estimate suggested by Pennington (1985) (part 1 above) is investigated in the case where the food-intake rate is constant and stomach contents are only sampled twice. The effects of the time elapsed between the sampling points and differences in evacuation rate on bias are examined. Further (part 2 above), an analytical expression for calculating an unbiased estimate of food intake directly from a pooled stomach sample is presented. Lastly (part 3 above), a new estimate of the intake of individual prey types is suggested. The estimate takes the distribution and evacuation of individual prey types into account as well as the effect of other food in the stomach on the evacuation rate. The bias introduced by ignoring these factors is analysed and exemplified by estimating the intake of English whiting (*Merlangius merlangus*) 25–30 cm in length caught at five different locations in the North Sea.

Analytical considerations

Bias in Pennington's method

The purpose of this exercise was to determine if Pennington's method provides an unbiased estimate of food intake when applied to a predator feeding at a constant rate between two sampling points. This was done by solving eq. 1 using numerical integration in a standard mathematical package for a number of combinations of known stomach contents at $t = 0$, $S(0)$, food-intake rate C, time interval T, and evacuation rate. The coefficient α was fixed at 0.5, as this value is appropriate for a number of fish species (Jobling

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1981; Andersen 2001). The numerical integration resulted in an estimate of the stomach contents at time T, denoted $S(T)$ below. Using this result, the ability of Pennington's approximation to estimate food-intake rate was determined by calculating

$$C_P = \frac{\rho}{2}\left(\sqrt{S(0)} + \sqrt{S(T)}\right) + \frac{1}{T}(S(T) - S(0))$$

The relative bias was calculated as

$$\frac{C_P - C}{C}$$

for each combination of $S(0)$, C, ρ, and T. Relative bias was estimated by for all combinations of $S(0)$ and C in the interval from 0.05 to 3.05 g. Evacuation rates of 0.04, 0.14, and 0.24 were used with a fixed time interval of 4 h to examine the effect of evacuation rate. These rates correspond to evacuation of slowly evacuated prey from whiting stomachs at 8 °C and of rapidly evacuated prey at 16 °C. The effect of the length of the interval between consecutive samples was examined by fixing the evacuation rate at 0.24 and estimating the difference for T = 1 h, T = 4 h, and T = 10 h.

Calculating an unbiased estimate of consumption from samples with a constant coefficient of variation

To obtain an estimate of $E(S_i^\alpha)$, $\hat{S}_i$, directly from the observed average stomach contents in a pooled sample, Andersen (2001) proposed a correction factor, κ, which should be multiplied to $(\bar{S}_i)^\alpha$:

$$\hat{S}_i^\alpha = \kappa(\bar{S}_i)^\alpha$$

The correction factor was calculated from a data set consisting of individually analysed saithe (*Pollachius virens*) stomachs by comparing $(\bar{S}_i)^\alpha$ and $\overline{S_i^\alpha}$. The factor was then applied to a data set consisting mainly of samples where the contents of the non-empty stomachs were pooled prior to weighing. However, if the distribution of the stomach contents follows a known distribution, the correction factor can be calculated analytically as well as numerically.

The distribution of the weights of the stomach contents has previously been shown to be reasonably described by delta distributions based on lognormal or gamma distributions (Aitchison and Brown 1957; Stefánsson and Pálsson 1997):

$$P(S = 0) = 1 - p$$

$$P(S \le s) = p\,\mathrm{LN}(s|\mu, \sigma^2), \quad S > 0$$

or

$$P(S = 0) = 1 - p$$

$$P(S \le s) = p\Gamma(s|k, \beta), \quad S > 0$$

where p is the probability that a stomach is not empty, $\mathrm{LN}(s|\mu, \sigma^2)$ and $\Gamma(s|k, \beta)$ are the distribution functions of a lognormal distribution with mean μ and variance σ^2 and a gamma distribution with shape parameter k and scale parameter β, respectively. In both cases, $E(S_i^\alpha)$ can be derived analytically and can be estimated by

$$\hat{S}_i^\alpha = \hat{p}_i\, e^{\alpha\hat{\sigma}_i^2(\alpha-1)/2}(\bar{x}_i)^\alpha \quad (3)$$

$$\hat{S}_i^\alpha = \hat{p}_i \frac{\Gamma(\hat{k}_i + \alpha)}{\Gamma(\hat{k}_i)\hat{k}_i^\alpha}(\bar{x}_i)^\alpha \quad (4)$$

(Appendix A) in the lognormal and gamma case, respectively, where $\bar{x}_i$ denotes the average contents of non-empty stomachs in the sample. If the coefficient of variation (CV) for the non-empty stomachs is constant, eqs. 3 and 4 reduce to

$$\hat{S}_i^\alpha = \hat{p}_i\, e^{\alpha\hat{\sigma}^2(\alpha-1)/2}(\bar{x}_i)^\alpha \quad (5)$$

$$\hat{S}_i^\alpha = \hat{p}_i \frac{\Gamma(\hat{k} + \alpha)}{\Gamma(\hat{k})\hat{k}^\alpha}(\bar{x}_i)^\alpha \quad (6)$$

in the lognormal and gamma case, respectively, where σ and k are the common parameters for all samples. These parameters must be estimated from stomachs analysed individually, preferably from a subsample of the data set. In the case of a constant CV, eq. 5 or 6 can then be used to estimate $E(S_i^\alpha)$ for samples where the contents of non-empty stomachs were pooled prior to weighing; $\hat{p}_i$ and $\bar{x}_i$ can be calculated from these samples, whereas σ_i or k_i is unknown. If a subsample is not available for estimating σ or k, the parameters could be estimated from a data set consisting of individually analysed stomachs sampled under similar conditions. It is readily apparent that the precision of eqs. 5 and 6 will depend on the degree to which the assumption of a constant CV is fulfilled. This assumption appeared to hold true for the contents of non-empty cod stomachs collected in restricted areas (Stefánsson and Pálsson 1997; Adlerstein and Welleman 2000). However, if an attempt is made to estimate the CV from other data sets, the validity of the assumption should be investigated further.

In addition to using eqs. 5 and 6 to correct for pooled samples, they can be applied in cases where stomachs were analysed individually but a common CV could be estimated for the entire data set. In this case, the simple average $\overline{S_i^\alpha}$ does not provide the best estimate of $E(S_i^\alpha)$, and eqs. 5 and 6 provide more precise estimates, as the information inherent in a common CV is taken into account.

The estimation of the variance of the estimated consumption has been described by Pennington (1985). However, if eq. 5 or 6 is used to estimate $E(S_i^\alpha)$, the equations do not apply. An estimate of the limits of the variance of consumption is given for both distributions (Appendix B).

The total consumption rate can be estimated by inserting eq. 5 or 6 into eq. 2. In the case of gamma-distributed non-zero stomach contents, α = 0.5, and a constant CV, the estimation is demonstrated in Table 1 (C_1). The corresponding estimate derived by simply substituting $(\bar{S}_i)^\alpha$ for $E(S_i^\alpha)$ is also given (C_2; Table 1). Assuming that $\bar{S}(0) = \bar{S}(T)$, the bias introduced by using C_2 rather than C_1 is

$$\frac{C_2 - C_1}{C_1} = \frac{\sum_{i=1}^{N}(p_i\bar{x}_i)^\alpha}{e^{\alpha\sigma^2(\alpha-1)/2}\sum_{i=1}^{N} p_i(\bar{x}_i)^\alpha} - 1 \quad (7)$$

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Table 1. Estimates of average hourly consumption.

Estimate	Calculation
C_1	$\frac{\rho_{tot}}{N}\sum_{i=1}^{N}\hat{p}_i\frac{\Gamma(\hat{k}_{tot}+1/2)}{\Gamma(\hat{k}_{tot})\sqrt{\hat{k}_{tot}}}\sqrt{\hat{X}_i}+\frac{1}{T}[\hat{p}(T)\hat{X}(T)-\hat{p}(0)\hat{X}(0))]$
C_2	$\frac{\rho_{tot}}{N}\sum_{i=1}^{N}\sqrt{\hat{p}_i\hat{X}_i}+\frac{1}{T}[\hat{p}(T)\hat{X}(T)-\hat{p}(0)\hat{X}(0)]$
$C_{3,j}$	$\frac{1}{N}\sum_{i=1}^{N}\sqrt{\hat{\pi}_{j,i}}(\hat{\pi}_{j,i}\rho_j+(1-\hat{\pi}_{j,i})\rho_{res})\hat{p}_{j,i}\frac{\Gamma(\hat{k}_j+1/2)}{\Gamma(\hat{k}_j)\sqrt{\hat{k}_j}}\sqrt{\hat{X}_{j,i}}+\frac{1}{T}[\hat{p}_j(T)\hat{X}_j(T)-\hat{p}_j(0)\hat{X}_j(0)]$
$C_{4,j}$	$\hat{\pi}_{tot,j}C_1$
$C_{5,j}$	$\frac{\rho_j}{N}\sum_{i=1}^{N}\hat{p}_{j,i}\frac{\Gamma(\hat{k}_j+1/2)}{\Gamma(\hat{k}_j)\sqrt{\hat{k}_j}}\sqrt{\hat{X}_{j,i}}+\frac{1}{T}[\hat{p}_j(T)\hat{X}_j(T)-\hat{p}_j(0)\hat{X}_j(0)]$

Note: See the text for an expanation of the notation.

in the case of lognormal-distributed contents of non-empty stomachs and

$$\frac{C_2-C_1}{C_1}=\frac{\sum_{i=1}^{N}(p_i\bar{x}_i)^{\alpha}}{\frac{\Gamma(k+\alpha)}{\Gamma(k)k^{\alpha}}\sum_{i=1}^{N}p_i(\bar{x}_i)^{\alpha}}-1 \quad (8)$$

in the case of gamma-distributed contents of non-empty stomachs. This bias was estimated for a number of combinations of p and k and p and σ, respectively, while α was held constant at 0.5; p was assumed to remain constant over the sampling period.

Estimation of the intake of individual prey types

Traditional estimates of prey-specific intake are based either on the distribution of total intake among prey types according to their weight percentages (Armstrong 1982; Hawkins et al. 1985; Persson and Hansson 1999) or on calculation of the intake of each prey type separately. The former method assumes that all prey types are present in the average amount in all stomachs and evacuated at the same rate, while the assumption of the latter method is that other food in the stomach has no effect on evacuation. Neither of these assumptions is generally met (Dos Santos and Jobling 1995; Stefánsson and Pálsson 1997; Andersen 2001). There is thus a need for an estimate that takes into account both the effect of the distribution and evacuation of individual prey types and the effect of other food in the stomach on evacuation.

A new estimate

The presence of other food in the stomach alters the evacuation of each particular prey type (Jones 1974), and the rate of evacuation of the total stomach contents lies within the range of rates for the individual prey types in the stomach (Andersen 2001). The proportion of all evacuated material that consists of a particular prey type is identical with the weight proportion of the prey in the stomach contents (Andersen 2001). In this study it will be assumed that the rate of evacuation of the total contents can be approximated by the average of the evacuation rates of the separate prey types weighted by the weight of each prey type, though this may not be entirely correct (Andersen 2001). The differential equation describing the evacuation of the total stomach contents is then

$$\frac{dS_{tot}(t)}{dt}=-(\rho_j\pi_j(t)+(1-\pi_j(t))\rho_{res})S_{tot}^{\alpha}(t)+C_{tot}(t) \quad (9)$$

where $\pi_j(t)$ denotes the weight proportion of prey species j in the stomach at time t and S_{tot} denotes the total weight of food in the stomach. The parameters all refer to stomachs where prey type j is present, as the prey obviously does not affect evacuation in stomachs where it is absent; ρ_j is the evacuation rate of the prey in question and ρ_{res} is the evacuation rate of the other stomach contents (i.e., excluding prey type j). The differential equation describing the evacuation of prey type j becomes

$$\frac{dS_j(t)}{dt}=-\pi_j(t)[\rho_j\pi_j(t)+(1-\pi_j(t))\rho_{res}]S_{tot}^{\alpha}(t)+C_j(t) \quad (10)$$

Recall that the proportion of the total amount of food evacuated that consists of prey type j is identical with the weight proportion of j in the stomach where the prey type occurs, i.e.,

$$\pi_j(t)=\frac{S_j(t)}{S_{tot}(t)}$$

Equation 10 can also be written as

$$\frac{dS_j(t)}{dt}=-\pi_j^{1-\alpha}(t)[\pi_j(t)\rho_j+(1-\pi_j(t))\rho_{res}]\times S_j^{\alpha}(t)+C_j(t) \quad (11)$$

$\bar{C}_j$ can then be estimated by

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$$C_{3,j} = \frac{1}{N}\sum_{i=1}^{N} \pi_{j,i}^{1-\alpha}[\pi_{j,i}\rho_j + (1-\pi_{j,i})\rho_{\text{res}}] \times E(S_{j,i}^{\alpha}) + \frac{1}{T}(E(S_j(T)) - E(S_j(0))) \quad (12)$$

(Table 1). The subscript 3 distinguishes this estimate from those obtained by other methods. This new estimate thus takes into account both the uneven distribution of prey types in the stomachs and the effect of other stomach contents on the evacuation of the particular prey type. $E(S_i^{\alpha})$ can be estimated directly from individually analysed stomachs or using eq. 5 or 6. In the latter case, an estimate of the occurrence of the prey type, p_j, is required as well as an estimate of σ_j or k_j.

Comparison of the new estimate with traditional methods

A number of authors estimate the intake of individual prey types by distributing the total intake according to the weight percentages of the different prey types (Armstrong 1982; Hawkins et al. 1985; Seyhan and Grove 1998):

$$C_{4,j} = \pi_{\text{tot},j} C_1 \quad (13)$$

where $\pi_{\text{tot},j}$ is the weight percentage of prey type j in the contents of all stomachs and C_1 is the total food intake. This estimate is biased even in the case where both the evacuation rates and the CVs of the weight of the stomach contents are identical for prey type j and the total stomach contents (as shown in Appendix C). The effect on the bias of the proportion of stomachs that contain the particular prey type, p_j, relative to the probability that a stomach contains food of any kind, p_{tot}, and the effect of the ratio between the weight proportion of the prey type in all stomachs, $\pi_{\text{tot},j}$, and in stomachs where the prey type occurred, π_j, was investigated in the special case where $\rho_j = \rho_{\text{tot}}$, $k_j = k_{\text{tot}}$, no change occurred in the average stomach contents over the sampling period, and $\pi_{\text{tot},j}$ and π_j remain constant over the period. In this case, the relative bias is reduced to

$$\frac{C_4 - C_3}{C_3} = \left(\frac{\pi_{\text{tot},j}}{\pi_j}\frac{p_{\text{tot}}}{p_j}\right)^{1-\alpha} - 1 \quad (14)$$

(Appendix C). This bias was plotted as a function of the ratios $\frac{\pi_{\text{tot},j}}{\pi_j}$ and $\frac{p_{\text{tot}}}{p_j}$ for $\alpha = 0.5$. The plot illustrates the bias introduced even in the case where the evacuation rate does not vary among prey types.

An alternative procedure is to calculate the intake of each prey type separately (Armstrong 1982; Patterson 1985). The assumption is that other food in the stomach has no effect on evacuation. The differential equation describing the stomach content then becomes

$$\frac{dS_j(t)}{dt} = -\rho_j S_j^{\alpha}(t) + C_j(t)$$

and food intake rate can be calculated as

$$C_{5,j} = \frac{\rho_j}{N}\sum_{i=1}^{N} E(S_{j,i}^{\alpha}) + \frac{1}{T}[E(S_j(T)) - E(S_j(0))] \quad (15)$$

The formula is thus identical with eq. 2, except that it refers only to the weight and evacuation rate of prey type j. If π_j remains constant over the period, the bias introduced by this method reduces to

$$\frac{C_{5,j} - C_{3,j}}{C_{3,j}} = \frac{\rho_j}{(\pi_j\rho_j + (1-\pi_j)\rho_{\text{res}})\pi_j^{1-\alpha}} - 1 \quad (16)$$

(Appendix C). This bias was plotted as a function of ρ_{res}/ρ_j and π_j.

Materials and methods

Data set

The data were obtained from whiting 25–30 cm in length gathered around the clock at five locations in the North Sea. Trawl hauls were performed at 4-h intervals for a total of 48–72 h. The sampling procedure has been described in detail elsewhere (Rindorf 2002) as has the calculation of the occurrence and weight of different prey types (Rindorf 2003, 2004). The total weight present in non-empty stomachs and the occurrence of non-empty stomachs were analysed using models similar to the analyses of total weight and occurrence of a single prey type described in Rindorf (2003, 2004). Briefly, the effect of time of day, time elapsed since the first sample was taken, and location on occurrence and mean weight of single prey and total stomach contents was analysed. If a factor was found to have no significant effect on weight or occurrence, samples were joined prior to estimation of mean weight or occurrence, respectively. This procedure was used to minimize random errors in the estimates. The weight of both single prey and total stomach contents in stomachs where this exceeded zero was assumed to be gamma-distributed with a constant CV.

The average hourly food intake was calculated from the estimated average stomach contents in non-empty stomachs, the estimated common shape parameter, k, and the occurrence of non-empty stomachs, p. The estimation of these parameters was described in detail in Rindorf (2003, 2004).

The bias introduced by not correcting for the inequality between $(\bar{S}_i)^{\alpha}$ and $\overline{S_i^{\alpha}}$ was examined by estimating the total consumption as C_1 (correcting) and C_2 (not correcting) (Table 1). The bias introduced by the two simplified methods for calculating intake of individual prey was investigated by estimating the intake of three different prey types as C_3, C_4, and C_5 (Table 1). Benthic invertebrates were chosen as an example of a prey that was generally small, occurred frequently in the stomachs, and had a high evacuation rate. Atlantic herring (*Clupea harengus*) was chosen as an infrequent large prey with an evacuation rate close to that of the total stomach contents. Finally, crabs were selected to exemplify prey that occurred frequently and possess a hard exoskeleton and thus are evacuated at a low rate.

Estimation of evacuation rates, $\pi_{\text{tot},j}$ and π_j

The evacuation rate is dependent on both temperature and predator and prey characteristics. Additionally, the evacuation rate has been found to vary with meal size in some cases (Garber 1983; Elliott 1991; Temming and Andersen 1994), but as this effect has been shown to be insignificant in other cases (Bromley 1987; Andersen 1998), it was not

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considered in this study. The estimation of evacuation rates is described in detail in Appendix D and the resulting values are given in Table 2. The percentage of the total stomach contents that consisted of each prey type, $\pi_{tot,j}$, was calculated for each location and time of day. In addition, the percentage of the stomach contents that was made up by prey type j in the stomachs where it occurred, π_j, was calculated for each prey type, location, and time of day. Therefore, π_j was assumed to be constant and statistically independent of S_j for a given time of day and location. As changes in S_j occurred through changes in p_j rather than in $\bar{x}_j$ (though trends in the former occurred over the sampling period, no trends were detected in the latter (Rindorf 2003, 2004)), this assumption appeared to be justified.

Results

Bias in Pennington's method

The bias was small for short intervals and in no case exceeded 3% when the time interval was 1 h. However, increasing the interval led to large differences when the stomach contents or intake was low (Fig. 1). The bias rose to +34% at a 10-h sampling interval and a very low food-intake rate. The difference between the two estimates increased with evacuation rate, being virtually zero for an evacuation rate of 0.04 (maximum bias –1.4%), rising to a maximum of 8% during a 4-h interval at an evacuation rate of 0.24 (Fig. 1). For the large majority of combinations of C and $S(0)$, the difference was less than 5% if the sampling interval was 4 h or less.

Calculating an unbiased estimate of consumption from samples with a constant CV

Substantial bias was introduced by estimating $E(\sqrt{S_i})$ as $\sqrt{\bar{S}_i}$. The bias increased dramatically as a function of σ^2 and k^{-1} in the lognormal and gamma-distributed data, respectively (Fig. 2). Empty stomachs further increased the bias, leading to a maximum of several hundred percent in both distributions. The consumption calculated from the whiting data was increased by 51%, on average, when $E(\sqrt{S_i})$ was estimated as $\sqrt{\bar{S}_i}$ (Table 3). The common shape parameter, k, of the distribution of the total stomach contents was 0.269 and the proportion of stomachs that contained food was around 0.8 (Table 3).

Estimation of prey-specific intake

Distributing the total intake among prey types according to weight percentage seriously biased the estimates (Fig. 3). The area in which the bias was less than 20% constituted a very small part of the possible parameter combinations. If the prey type was found in most of the non-empty stomachs, the bias was generally negative. As the occurrence of the prey type in the non-empty stomachs decreased, a positive bias became more likely. The data example demonstrated this, as the estimate $C_{4,j}$ was lower than $C_{3,j}$ when the prey type occurred frequently in the stomachs (Table 4). The estimate $C_{4,j}$ exceeded $C_{3,j}$ at low occurrences, particularly where the prey type constituted less than three-quarters of the stomach contents in the stomachs where it occurred. $C_{4,j}$ was biased by –36% and –44%, on average, for benthic invertebrates and crabs, respectively, when intake exceeded 3 mg·h^{-1}. Herring estimated by $C_{4,j}$ was positively biased at location 3, though the occurrence was similar to that at location 1, where a slight negative bias was found. This was caused by a higher mean weight of herring at location 3 (Appendix C).

Table 2. Rates of evacuation of total stomach contents and individual prey types in locations 1–5.

	1	2	3	4	5
Temperature (°C)	16.7	13.7	10.6	8.2	8.6
ρ_{tot}	0.120	0.098	0.087	0.075	0.077
$\rho_{benthic\ invertebrates}$	0.183	0.145	0.113	0.094	0.097
$\rho_{herring}$	0.160	0.127	0.099	0.082	0.085
ρ_{crabs}	0.091	0.072	0.056	0.047	0.048

Failing to include the effect on evacuation of other food in the stomach introduced a positive bias in consumption. When the prey type in question constituted half the weight of the stomach contents, consumption was estimated to be approximately 140% of the true consumption even when the rate of evacuation of the prey was identical with that of the total stomach contents (Fig. 4). However, as the prey type often constituted most of the stomach contents when it was present in the whiting stomachs, the positive bias in estimated consumption was only 26%, 9%, and 10%, on average, for benthic invertebrates, herring, and crabs, respectively (Table 4). The relative degree of bias in the three groups reflected that expected from the evacuation rate of the prey and the proportion of the stomach contents made up by this prey type (Table 4).

Discussion

The bias introduced by using Pennington's method to estimate a constant food-intake rate was very small for sampling intervals of 4 h or less. In contrast to this, simplifying the calculation of input data and making unwarranted assumptions regarding the distribution of prey in the stomachs led to serious bias in the estimated food intake.

The estimation of $E(S_i^\alpha)$ by $(\bar{S}_i)^\alpha$ led to substantial overestimation of consumption. This effect was also noted by Ursin et al. (1985), Dos Santos and Jobling (1995), and Andersen (2001) using different methods. The latter author estimated correction factors that could be applied to eliminate bias from data obtained from individually analysed stomachs. Though individual stomachs are also needed to estimate the parameters in the correction suggested here, the two methods differ in their underlying assumptions. Whereas Andersen's (2001) correction factor assumes a constant CV for the contents of non-empty stomachs as well as a constant proportion of empty stomachs, the present method allows these assumptions to be made independently of each other, thus ensuring a minimum number of fixed parameters. Further, the method presented here provides a lower limit of the variance of the estimated consumption. The distribution and CV can be estimated from a subsample of the data set, and the remaining data can then be analysed by counting the empty and non-empty stomachs and recording the mean weight of

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Fig. 1. The bias introduced by using Pennington's method (eq. 2) to estimate a constant rate of food intake, showing the effect of evacuation rate at a 4-h sampling interval (*a* and *b*) and effect of the interval between sampling points at an evacuation rate of 0.24 (*b* and *c*). (*a*) ρ = 0.14 and T = 4; (*b*) ρ = 0.24 and T = 4; (*c*) ρ = 0.24 and T = 10. Relative differences are 0%–2%, 2%–4%, 4%–6%, 6%–8%, and above 8%, which are indicated by progressively darker shading. Positive differences are indicated by grids. The highest observed bias was 34% and the lowest was –15%.

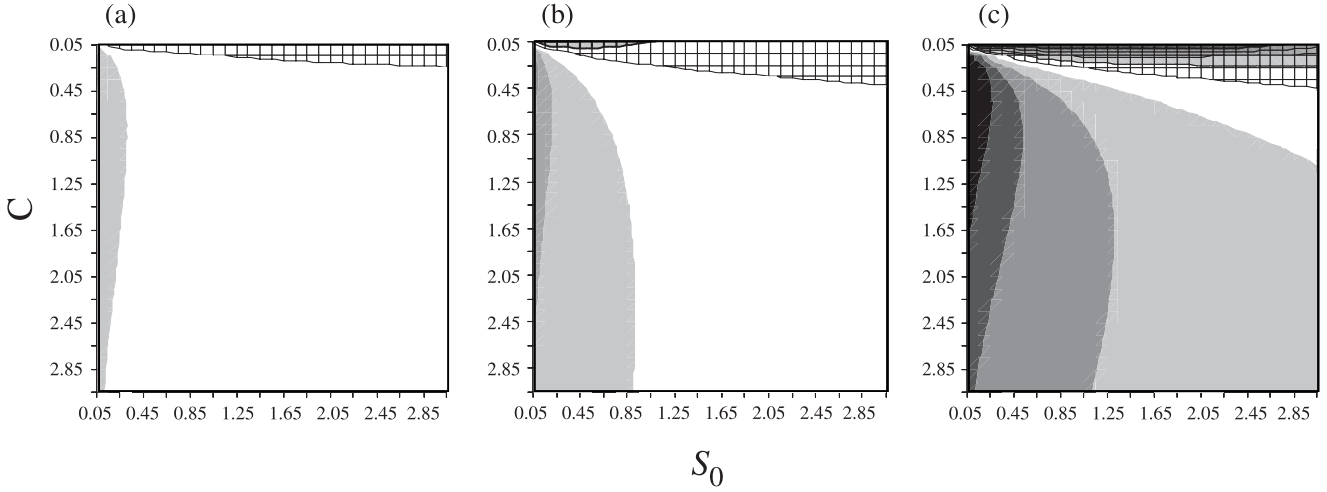
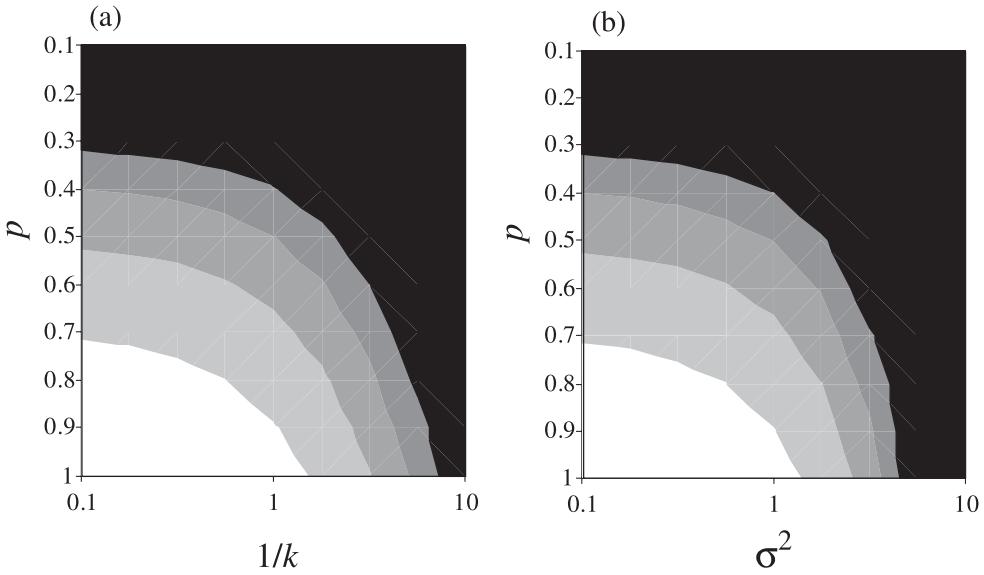


Fig. 2. Bias introduced by calculating $\sqrt{\hat{S}_i}$ as $\sqrt{\bar{S}_i}$ when X is (*a*) gamma-distributed and (*b*) lognormal-distributed as a function of the proportion of stomachs that contain food, p, and 1 divided by the shape parameter of the gamma distribution, k^{-1}, and the variance of the lognormal distribution, σ^2, respectively. The degrees of bias are 0%–20%, 20%–40%, 40%–60%, 60%–80%, and above 80%, which are indicated by progressively darker shading. The maximum recorded bias was 540% in the gamma-distributed case and 1000% in the lognormal-distributed case.



the contents of the non-empty stomachs. This sampling procedure has traditionally been followed in a number of stomach-sampling projects (International Council for the Exploration of the Sea 1988, 1991). The correction requires a constant CV, an assumption that may not hold if extensive areas or time scales are considered. There does, however, appear to be some evidence of constancy, as Stefánsson and Pálsson (1997) and Adlerstein and Welleman (2000) found evidence of a common CV for the contents of non-empty stomachs of Icelandic and North Sea cod, respectively. Nevertheless, the assumption is of crucial importance to the validity of the correction, and should be considered in detail before attempting to correct large numbers of samples.

Simply excluding empty stomachs leads to an overestimate of food intake, as the fish that were caught in the interval between the complete evacuation of their last meal and their successful capture of the next prey are excluded. Empty stomachs should only be excluded when there is biological evidence that these fish were indeed not part of the feeding population. In this study, there was no evidence that some predators may have been fasting. However, if some predators appeared to have decreased their food intake, because of

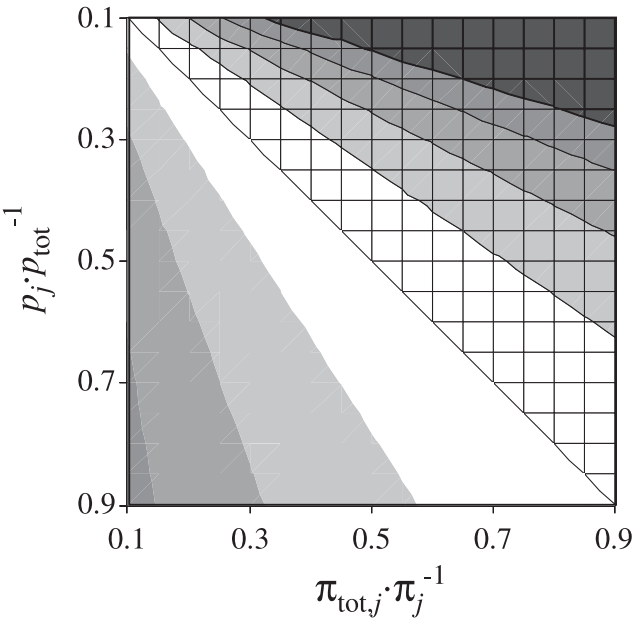
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Table 3. Comparison of average food intake of whiting ($mg \cdot h^{-1}$) estimated by correcting for the fact that $E(S_i^{\alpha}) \neq (E(S_i))^{\alpha}$ (C_1) and by assuming $E(S_i^{\alpha})$ to be equal to $(E(S_i))^{\alpha}$ (C_2).

Location	C_1	C_2	$(C_2 - C_1)/C_2$	p^*
1	132	187	0.42	0.826
2	66	100	0.52	0.779
3	112	157	0.40	0.844
4	30	49	0.63	0.746
5	69	109	0.58	0.634

*The proportion of stomachs that contained food.

Fig. 3. Bias introduced by assuming that all prey types occur in all stomachs in the average amount ($(C_{4,j} - C_{3,j}) \cdot C_{3,j}^{-1}$) as a function of the proportion of stomachs with food that contain prey type j, $p_j \cdot p_{tot}^{-1}$, and the ratio of the proportion of the prey type in the total stomach contents to the contents of the stomachs in which the prey type is found, $\pi_{tot,j} \cdot \pi_j^{-1}$. The degrees of bias are 0%–20%, 20%–40%, 40%–60%, 60%–80%, and above 80%, which are indicated by progressively darker shading. Positive differences are indicated by grids. The highest observed bias was 200%.



spawning, for example, the biologically most appropriate estimate of food intake would appear to be separate estimates for spawning and nonspawning fish; hence, empty and nonempty stomachs of spawning fish are considered together.

The new method for calculating the consumption of individual prey types yielded greatly improved estimates. Compared with this method, consumption of the three prey types considered was seriously biased by distributing the total consumption among prey types using weight percentages. Though the bias was generally negative, a positive bias was found for prey that occurred infrequently or constituted a minor part of the stomach contents when present. In general, larger fish prey occur in only a minor percentage of the stomachs collected (Albert 1995; Pillar and Barange 1997)

Table 4. Consumption ($mg \cdot h^{-1}$) of individual prey types at locations 1–5 estimated by three different methods, correcting for the unequal distribution of prey types and the effect of other food on evacuation ($C_{3,j}$), applying weight percentages to total consumption ($C_{4,j}$), and correcting for the unequal distribution of prey but neglecting the effect of other prey on evacuation ($C_{5,j}$).

	1	2	3	4	5
Benthic invertebrates					
Average π_j	0.72	0.81	0.67	0.73	0.68
Average p	0.23	0.39	0.40	0.08	0.02
Average X	0.74	0.74	0.74	0.74	0.74
$C_{3,j}$	17.2	37.0	24.2	3.7	1.3
$C_{4,j}$	7.1	32.6	16.1	2.3	3.2
$C_{5,j}$	22.4	41.6	31.1	4.7	1.7
Herring					
Average π_j	0.96	0.84	0.92	1.00	0.98
Average p	0.04	0.03	0.04	0.001	0.01
Average X	6.7	6.6	14.1	3.4	5.8
$C_{3,j}$	36.6	2.0	14.9	0.2	1.6
$C_{4,j}$	30.4	2.3	31.1	0.5	2.8
$C_{5,j}$	37.8	2.7	15.6	0.2	1.6
Crabs					
Average π_j	0.50	0.57	0.48	0.60	0.50
Average p	0.23	0.18	0.34	0.16	0.003
Average X	0.29	0.73	0.29	0.37	0.22
$C_{3,j}$	10.3	15.2	8.6	3.3	0.07
$C_{4,j}$	6.6	2.9	7.3	5.5	0.13
$C_{5,j}$	12.0	16.4	9.6	3.4	0.08

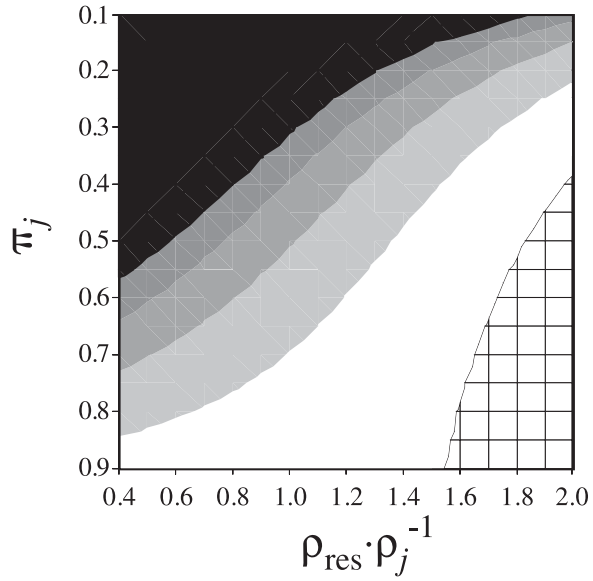
and so consumption of fish is likely to be overestimated, whereas that of invertebrates will be underestimated. This conclusion holds even when no difference in evacuation rate exists between the two prey types. Fish in which the evacuation of the stomach contents is exponential ($\alpha = 1$) represent a special case in which the error in distributing the total consumption according to weight percentages depends solely on the difference in evacuation rate between prey types (Appendix C).

The effect of other prey types on evacuation and hence consumption was minor in most cases. Only the intake of benthic invertebrates was biased to any extent by this assumption, and if simplifications of the calculations are to be made, making the assumption that each prey type is evacuated separately is far superior to distributing consumption according to weight percentages. Unfortunately, in studies calculating the intake of single prey types by this method, $E(S_i^{\alpha})$ is frequently estimated as $(\bar{S}_i)^{\alpha}$ (Armstrong 1982; Patterson 1985) and the results are therefore positively biased.

Estimating the intake of each prey type in the data set and summing these values to obtain the total intake led to estimates of total intake of 62%–131% of C_1 in the data analysed here. This difference may be the result of a combination of random errors and a tendency to overestimate the mean weight of highly digested prey, as prey in the last stages of digestion are not identified. However, the dif-

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Fig. 4. Bias introduced by assuming that no other food is present in the stomach ($(C_{5,j} - C_{3,j}) \cdot C_{3,j}^{-1}$) as a function of the proportion of the stomach contents that consists of the given prey type, π_j, and the ratio of the evacuation rate of this prey type to that of the other stomach contents, $\rho_{res} \cdot \rho_j^{-1}$. The degrees of bias are 0%–20%, 20%–40%, 40%–60%, 60%–80%, and above 80%, which are indicated by progressively darker shading. The maximum recorded bias was 475%.



ference may also reflect a biologically more sensible method of calculation. Calculating an average rate of evacuation of the total stomach contents fails to account for any correlation between evacuation rate and weight of the stomach contents. Such a correlation may arise if the stomachs contain, for example, either small crustaceans (low evacuation rate) or large lean fish (high evacuation rate).

In addition to determining the effect of using different methods to estimate consumption, the present study demonstrates the importance of collecting samples throughout the diel cycle: the average intake of crabs estimated from daytime samples alone was only 25%–50% of that estimated from nighttime samples in the data at hand. This was connected to the primarily nocturnal feeding on crabs (Rindorf 2003). Sampling during the daytime only would therefore lead to gross underestimation of the total intake of this prey type.

The new method for calculating the intake of individual prey types has the advantage of incorporating the known sources of bias in estimates of food intake while allowing for change in the average stomach contents during the sampling period. A major disadvantage, however, is the expense of collecting detailed information on the contents of individual stomachs. If the CV is constant for each prey type, the required information could be limited to the average weight, the CV of this, and the occurrence of each prey type. The analyses of the stomachs would then not require the weighing of prey from each stomach individually, except for the subsample necessary to estimate the correction factor. The effect of other food in the stomach on estimated consumption appears to be minor in most cases.

In conclusion, the estimation of $E(S_i^\alpha)$ as $(\bar{S}_i)^\alpha$ seriously biases the calculation of the food intake of fish. The bias increases with the variance of the contents of the non-empty stomachs and rises dramatically if the proportion of empty stomachs is high. Therefore, if data or CVs from single stomachs are not available, $E(S_i^\alpha)$ should at the very least be estimated as $p_i(\bar{X}_i)^\alpha$, as this would eliminate the severe effect of empty stomachs. Distributing the total food intake among different prey types according to their weight percentage produces large biases, which may be either positive or negative according to the distribution and evacuation rate of the prey. As this bias can take any value, it is even more dangerous than the bias in the simplified calculations, as the latter is at least known to be positive. The results shown here once again stress the need for information on stomach contents at the single-stomach level. If this type of information cannot be obtained, great care should be taken to minimize the otherwise substantial bias.

Acknowledgements

We thank two anonymous reviewers whose comments helped us to improve the manuscript. This study was partly funded by a grant from the Danish Ministry of Food, Agriculture and Fisheries to the project "Development of improved models of fisheries impact on marine fish stocks and ecosystems" (FIFOF-DFU-1) and by the European Union project "Development of structurally detailed statistically testable models of marine populations" (QLK5-CT1999-01609). The research was supported by the Scaling from Individuals to Populations (SLIP) Research School of the Danish Network for Fisheries and Aquaculture Research (www.fishnet.dk) financed by the Danish Ministry for Food, Agriculture and Fisheries and the Danish Agricultural and Veterinary Research Council.

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Appendix A. Estimation of $E(S_i^\alpha)$ from $E(S_i)$.

Let the distribution of the weights of the stomach contents, S, be described by a delta distribution based on lognormal or gamma distributions (Aitchison and Brown 1957; Stefánsson and Pálsson 1997):

$$P(S = 0) = 1 - p$$

$$P(S \le s) = p\mathrm{LN}(s|\mu, \sigma^2), \quad S > 0$$

or

$$P(S = 0) = 1 - p$$

$$P(S \le s) = p\Gamma(s|k, \beta), \quad S > 0$$

where p is the probability that a stomach is not empty, $\mathrm{LN}(s|\mu, \sigma^2)$ and $\Gamma(s|k, \beta)$ are the distribution functions of a lognormal distribution with mean μ and variance σ^2 and a gamma distribution with shape parameter k and scale parameter β, respectively. In both cases, $E(S_i^\alpha)$ can be derived analytically. Let the weight of the contents of a non-empty stomach be described by X, where X is lognormal distributed with mean μ and variance σ^2 of the log stomach contents:

$$X \in \mathrm{LN}(\mu, \sigma^2)$$

then

$$X^\alpha \in \mathrm{LN}(\alpha\mu, \alpha^2\sigma^2)$$

The relation between $(E(X))^\alpha$ and $E(X^\alpha)$ is

$$E(X^\alpha) = \mathrm{e}^{\alpha\sigma^2(\alpha-1)/2}(E(X))^\alpha$$

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and $E(S^\alpha)$ becomes

$$E(S^\alpha) = p e^{\alpha\sigma^2(\alpha-1)/2}(E(X))^\alpha$$

In the case of gamma distribution, $X \in \Gamma(k,\beta)$, $E(X^\alpha)$ is

$$E(X^\alpha) = \frac{\Gamma(k+\alpha)}{\Gamma(k)k^\alpha}(E(X))^\alpha$$

and

$$E(S^\alpha) = p\frac{\Gamma(k+\alpha)}{\Gamma(k)k^\alpha}(E(X))^\alpha$$

Hence $E(S_i^\alpha)$ can be estimated by

$$\hat{S}_i^\alpha = \hat{p}_i e^{\alpha\sigma_i^2(\alpha-1)/2}(\bar{x}_i)^\alpha$$

$$\hat{S}_i^\alpha = \hat{p}_i \frac{\Gamma(\hat{k}_i+\alpha)}{\Gamma(\hat{k}_i)\hat{k}_i^\alpha}(\bar{x}_i)^\alpha$$

in the lognormal- and gamma-distribution case, respectively.

Appendix B. Estimation of the variance of the estimated consumption.

The variance of the estimated consumption can be estimated as

$$V(\hat{C}) = \frac{\rho^2}{N^2}\sum_{i=1}^{N} V(\hat{S}_i^\alpha) + \frac{1}{T^2}[V(\hat{S}(T)) + V(\hat{S}(0))]$$

in the case where $S(0)$ and $S(T)$ are not included in the summation and

$$V(\hat{C}) \cong \frac{\rho}{N^2}\sum_{i=1}^{N} V(\hat{S}_i^\alpha) + \frac{1}{T^2}[V(\hat{S}(T)) + V(\hat{S}(0))] + \frac{\alpha\rho}{N\cdot T}[(\hat{S}(T))^{\alpha-1}V(\hat{S}(T)) - (\hat{S}(0))^{\alpha-1}V(\hat{S}(0))]$$

in the case where $S(0)$ and $S(T)$ are included in the summation. To calculate this variance, estimates of the variance of $\hat{S}(0)$, $\hat{S}(T)$, and $\hat{S}_i^\alpha$ are needed. If it is assumed that σ and k are constant and known to be without error, the variance of $\hat{S}_i^\alpha$ can be approximated by

$$V(\hat{S}_i^\alpha) \cong e^{\alpha\sigma^2(\alpha-1)}[V(\hat{p}_i)\,(\bar{X}_i)^{2\alpha} + \alpha(\bar{X}_i)^{\alpha-1}V(\bar{X}_i)\hat{p}_i^2 + \alpha(\bar{X}_i)^{\alpha-1}\,V(\bar{X}_i)\,V(\hat{p}_i)]$$

and

$$V(\hat{S}_i^\alpha) \cong \left(\frac{\Gamma(k+\alpha}{\Gamma(k)k^\alpha}\right)^2[V(\hat{p}_i)\,(\bar{X}_i)^{2\alpha} + \alpha(\bar{X}_i)^{\alpha-1}V(\bar{X}_i)\hat{p}_i^2 + \alpha(\bar{X}_i)^{\alpha-1}\,V(\bar{X}_i)\,V(\hat{p}_i)]$$

in the lognormal- and gamma-distributed case, respectively.

However, as the values of σ and k are unlikely to be known exactly, these estimates have to be considered a lower limit. An overestimate of the variance of $\hat{S}_i^\alpha$ can, by analogy with Lewy (1996), be calculated as

$$V(\hat{S}_i^\alpha) \cong \frac{V(S_i^\alpha)}{M_i} = \frac{1}{M_i}\hat{p}_i(e^{2\hat{\mu}_i+\hat{\sigma}_i^2})^\alpha \times (e^{\alpha\hat{\sigma}_i^2(2\alpha-1)} - \hat{p}_i e^{\alpha\hat{\sigma}_i^2(\alpha-1)})$$

and

$$V(\hat{S}_i^\alpha) \cong \frac{V(S_i^\alpha)}{M_i} = \frac{1}{M_i}\frac{\hat{p}_i(\hat{k}_i\hat{\beta}_i)^{2\alpha}}{\Gamma(\hat{k}_i)\hat{k}_i^{2\alpha}}\left(\Gamma(\hat{k}_i+2\alpha) - \hat{p}_i\frac{(\Gamma(\hat{k}_i+\alpha))^2}{\Gamma(\hat{k}_i)}\right)$$

in the lognormal- and gamma-distributed case, respectively, in the case where σ or k is estimated for each sample separately. If a common estimate of σ or k is available, this estimate will be an upper limit, as the gain in precision obtained by estimating a common CV is not accounted for.

The variance of $\hat{S}(0)$ and $\hat{S}(T)$ can be approximated by

$$V(\hat{S}_i) \cong V(\hat{p}_i)\,(\bar{X}_i)^2 + V(\bar{X}_i)\hat{p}_i^2 + V(\bar{X}_i)\,V(\hat{p}_i)$$

The approximations given above can only be used when the variance of $\bar{X}_i$ and $\hat{p}_i$ can be estimated. They can therefore not be applied to pooled stomach samples. In this case, the variance can be approximated by

$$V(\hat{S}_i^\alpha) \cong \frac{V(S_i^\alpha)}{M_i} = \frac{1}{M_i}(e^{\alpha\hat{\sigma}^2(2\alpha-1)} - \hat{p}_i e^{\alpha\hat{\sigma}^2(\alpha-1)})\hat{p}_i\bar{X}_i^{2\alpha}$$

and

$$V(\hat{S}_i^\alpha) \cong \frac{V(S_i^\alpha)}{M_i} = \frac{1}{M_i}\frac{1}{\Gamma(\hat{k})\hat{k}^{2\alpha}}\left(\Gamma(\hat{k}+2\alpha) - \hat{p}_i\frac{(\Gamma(\hat{k}+\alpha))^2}{\Gamma(\hat{k})}\right)\hat{p}_i\bar{X}_i^{2\alpha}$$

in the lognormal- and gamma-distributed case, respectively.

The variance of $\hat{S}(0)$ and $\hat{S}(T)$ can be approximated by

$$V(\hat{S}_i) \cong \frac{V(S_i)}{M_i} = \frac{1}{M_i}(e^{\hat{\sigma}^2} - \hat{p}_i)\hat{p}_i\bar{X}_i^2$$

and

$$V(\hat{S}_i) \cong \frac{V(S_i)}{M_i} = \frac{1}{M_i}\left(\frac{\hat{k}+1}{\hat{k}} - \hat{p}_i\right)\hat{p}_i\bar{X}_i^2$$

in the lognormal- and gamma-distributed case, respectively. The precision of these estimates depends entirely on the adequacy of the assumption of a constant CV and the precision of the estimates of σ and k. They are further lower limits, as the variance of $\hat{\sigma}$ and $\hat{k}$ is not taken into account.

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Appendix C. Estimation of bias in traditional methods for estimating the intake of individual prey types.

The relative difference between the intake of prey type j calculated by the method presented here ($C_{3,j}$) and by multiplying the weight percentage of prey type j in the total stomach contents, $\pi_{\text{tot},j}$ by the total intake ($C_{4,j}$) is

$$\text{(C1)}\quad \frac{C_{4,j}-C_{3,j}}{C_{3,j}} = \frac{\pi_{\text{tot},j}}{\pi_j^{1-\alpha}}\frac{\rho_{\text{tot}}}{(\pi_j\rho_j+(1-\pi_j)\rho_{\text{res}})}\frac{\sum_{i=1}^{N}E(S_{\text{tot}}^{\alpha})}{\sum_{i=1}^{N}E(S_j^{\alpha})}-1$$

where

$$\frac{\sum_{i=1}^{N}E(S_{\text{tot}}^{\alpha})}{\sum_{i=1}^{N}E(S_j^{\alpha})} = \frac{\frac{\Gamma(k_{\text{tot}}+\alpha)}{\Gamma(k_{\text{tot}})k_{\text{tot}}^{\alpha}}\sum_{i=1}^{N}p_{\text{tot},i}(E(X_{\text{tot},i}))^{\alpha}}{\frac{\Gamma(k_j+\alpha)}{\Gamma(k_j)k_j^{\alpha}}\sum_{i=1}^{N}p_{j,i}(E(X_{j,i}))^{\alpha}}$$

It is assumed that $\pi_{\text{tot},j}$ and π_j remain constant over the period and no change occurs in the average stomach contents over the sampling period. Note that in the special case $\alpha = 1$, the bias is entirely dependent on the difference in evacuation rate between prey.

If $\rho_{\text{tot}} = \rho_j = \rho_{\text{res}}$ and $k_{\text{tot}} = k_j$, eq. C1 can be reduced by inserting $E(X_{\text{tot}}) = \frac{p_j}{p_{\text{tot}}}\frac{E(X_j)}{\pi_{\text{tot},j}}$:

$$\text{(C2)}\quad \frac{C_{4,j}-C_{3,j}}{C_{3,j}} = \left(\frac{\pi_{\text{tot},j}}{\pi_j}\frac{p_{\text{tot}}}{p_j}\right)^{1-\alpha}-1$$

Once again it is evident that in the case of $\alpha = 1$, C_4 is only biased if evacuation rates vary among prey types.

The bias introduced by disregarding the effect of other food in the stomach on evacuation (i.e., estimating food intake as $C_{5,j}$) is

$$\text{(C3)}\quad \frac{C_{5,j}-C_{3,j}}{C_{3,j}} = \frac{\frac{\rho_j}{N}\sum_{i=1}^{N}S_i^{\alpha}}{\frac{(\pi_j\rho_j+(1-\pi_j)\rho_{\text{res}})\,\pi_j^{1-\alpha}}{N}\sum_{i=1}^{N}S_i^{\alpha}}-1 = \frac{\rho_j}{(\pi_j\rho_j+(1-\pi_j)\rho_{\text{res}})\pi_j^{1-\alpha}}-1$$

It is again assumed that π_j remains constant and no change in the average stomach contents occurs over the sampling period.

Appendix D. Estimation of evacuation rates.

Andersen (2001) showed that the rate of evacuation of individual prey types without an exoskeleton is well described by a function of energy density of the prey. If several prey types occurred together, the rate of evacuation of the total stomach contents could be estimated from the average energy density of the total stomach contents (Andersen 2001). The relationship between evacuation rate and energy density was used here to calculate the rate of evacuation of prey without an exoskeleton. Energy densities were assumed to be equivalent to values given by Andersen (2001) and Pedersen and Hislop (2001). The energy density of benthic invertebrates was assumed to be 3.7 $\text{kJ}\cdot\text{g}^{-1}$ based on the evacuation rate given by Rindorf (2004). The evacuation rate of the total stomach contents at each location was then calculated as a weighted average of the evacuation rate of prey without an exoskeleton (calculated from the average energy density of these prey) and the evacuation rate of prey with an exoskeleton (taken to be equal to the value given for common (brown) shrimp (*Crangon crangon*) in whiting 27.5 cm in length by Andersen (2001)):

$$\rho_{\text{tot}} = (0.0025\pi_{\text{exo}} + 0.152E^{-0.86}(1-\pi_{\text{exo}}))\,e^{0.078\ \text{temp}}$$

where π_{exo} is the proportion of the stomach contents that consists of prey with an exoskeleton at each location, $1-\pi_{\text{exo}}$ is the proportion of the stomach contents that consists of prey without an exoskeleton, E is average energy density of the prey without an exoskeleton, and "temp" refers to the ambient temperature at the location. This is a modified version of the equation given by Andersen (2001) for whiting 27.5 cm in length. The sizes and temperature-specific evacuation rates, ρ_{LT}, of the three prey types were 0.00042 for benthic invertebrates (Rindorf 2004), 0.00021 for crabs (value given for common (brown) shrimp in Andersen 1999), and 0.00037 for herring (derived from the relationship given by Andersen (2001), and an energy density of 4.3 $\text{kJ}\cdot\text{g}^{-1}$ (Pedersen and Hislop 2001)). With a predator length of 27.5 cm, the evacuation rates were 0.050 $e^{0.078\ \text{temp}}$, 0.025 $e^{0.078\ \text{temp}}$, and 0.032 $e^{0.078\ \text{temp}}$ for benthic invertebrates, herring, and crabs, respectively.

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